

**Q1.** A system has a transfer function,

$$G(s) = \frac{50}{s + 50}.$$

Find the time constant,  $T_c$ , settling time,  $T_s$ , and rise time,  $T_r$ .

**Q2.** Given the system of Figure 1, write the output,  $c(t)$ . Specify the forced and natural parts of the solution.

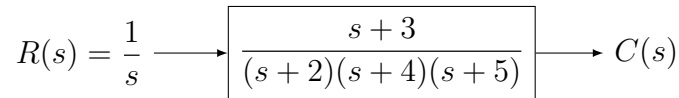


Figure 1: Transfer-function representation of the system.

**Q3.** A system has a transfer function,

$$G(s) = \frac{10(s+4)(s+6)}{(s+1)(s+7)(s+8)(s+10)}.$$

Determine the output,  $c(t)$ , if the input is a unit step.

**Q4.** For each of the system given below, determine the kind of response expected:

(a)  $G_1(s) = \frac{12}{s^2+8s+12}$

(b)  $G_2(s) = \frac{16}{s^2+8s+16}$

(c)  $G_3(s) = \frac{20}{s^2+8s+20}$

**Q5.** Given the system

$$G(s) = \frac{\frac{1}{J}}{s^2 + \frac{D}{J}s + \frac{K}{J}},$$

find  $J$  and  $D$  to yield 20% overshoot and a settling time of 2 seconds for a step input.

**Q6.** Given the pole plot shown in Figure 2, find

$$\zeta, \quad \omega_n, \quad T_p, \quad \%OS, \quad \text{and} \quad T_s.$$

**Q7.** Prove with the help of suitable argument and mathematical justification the behavior of the response of second-order system when poles move in s-plane as described in Figure 3.

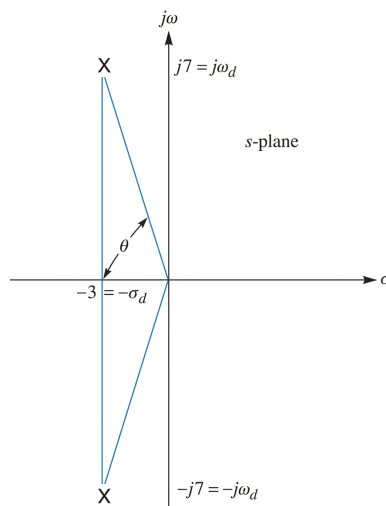


Figure 2: poles representation in s-plane

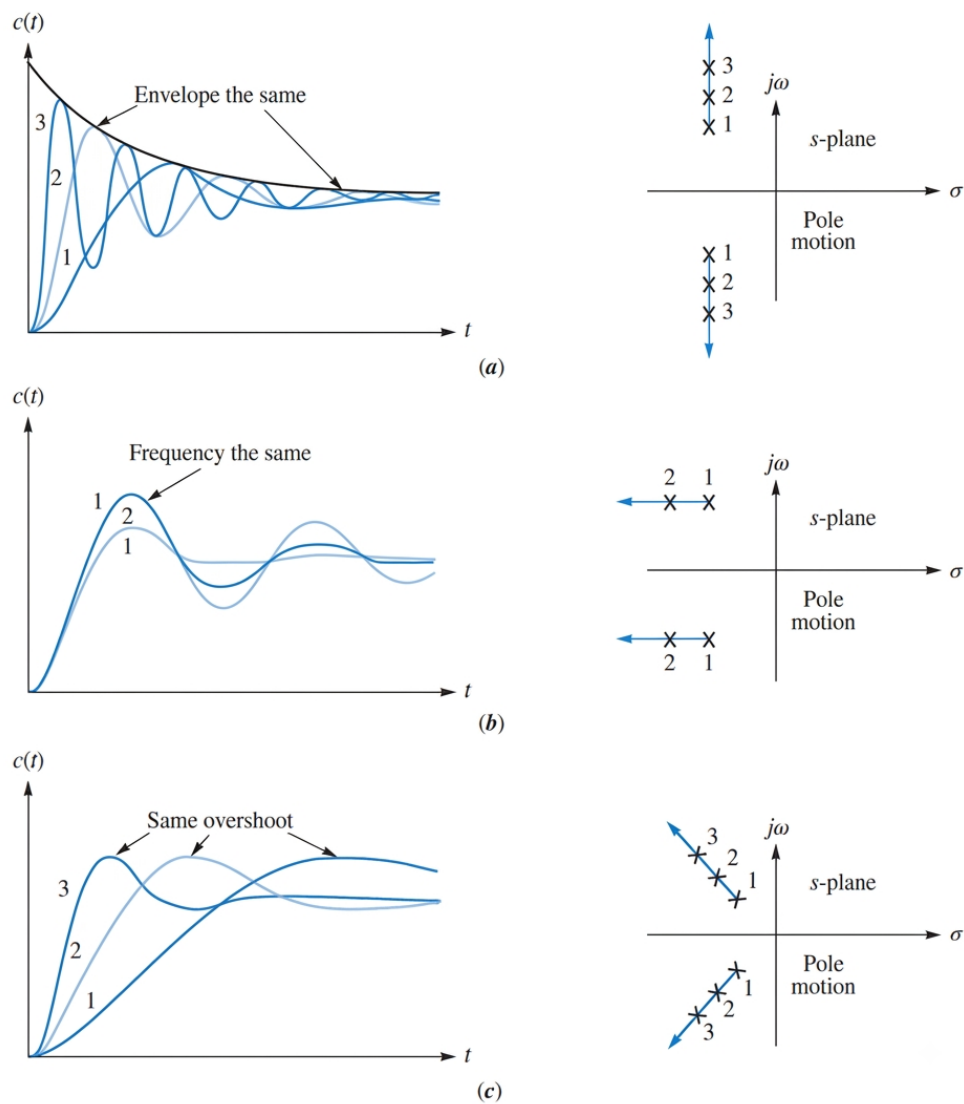


Figure 3: Step responses of second-order underdamped systems as poles move: a. with constant real part; b. with constant imaginary part; c. with constant damping ratio